
Real-Time ASL-to-English Translation Pipeline

Bridging Spatial-Temporal Tracking

Presenter

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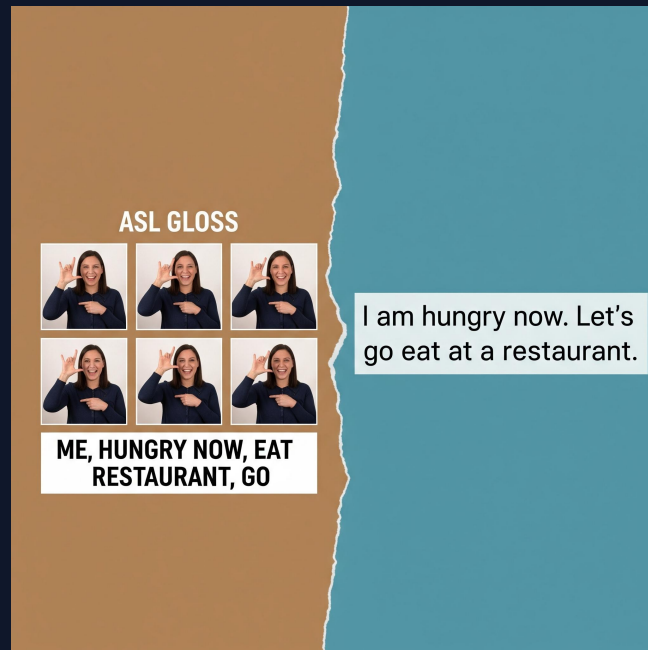
The Objective

The objective of this project is to develop an edge-efficient, real-time translation pipeline that bridges the gap between raw American Sign Language (ASL) gestures and grammatically correct English. By combining a lightweight Attention-LSTM neural network to track continuous temporal hand movements with a generative Large Language Model (LLM) to refine the resulting vocabulary, the system aims to seamlessly convert live spatial data into natural, conversational sentences.

Technical and Linguistic Challenges

Linguistic Disconnect

- ASL vs. English Grammar: ASL is a distinct language with its own syntax (Topic-Comment or Object-Subject-Verb). It lacks "to-be" verbs (is, am, are) and articles (the, a, an).
- Glossing: Standard vision models output "Gloss"—a literal, raw representation of signs (e.g., "ME PIZZA EAT")—which is unintelligible in professional or natural English contexts.



Technical and Linguistic Challenges

Sign Boundary Detection

- Continuous Fluidity: Unlike spoken language, which has distinct audio pauses, or written text with spaces, American Sign Language is continuously fluid.
- The Transition Challenge: Identifying precise "segmentation" points—the exact microsecond where one gesture ends and the transition to the next begins—is a significant temporal hurdle.

Proposed Model: **System** **Architecture**

Data Extraction

- Dataset: The translation pipeline was trained and validated on the [Google - Isolated Sign Language Recognition dataset](#), a large-scale repository comprising 100,000 ASL gesture sequences. Rather than relying on heavy raw video frames, the dataset consists entirely of pre-extracted MediaPipe geometric features—specifically spatial hand and body landmarks (x, y, z coordinates) captured over time.
- Continuous Data Pipeline: The system captures live webcam video and extracts 134 geometric hand landmarks per frame.
- Dimensionality Reduction: By focusing on geometric features, it successfully converts heavy, computationally expensive raw pixels into a lightweight, high-speed coordinate array.

Proposed Model: **System** **Architecture**

Attention-LSTM Hybrid Architecture

- Capturing Fluid Motion (The LSTM): Processes a continuous 30-frame sequence to track the full spatial trajectory of a sign, preventing the data loss associated with isolated frame analysis.
- Isolating the Apex (Temporal Attention): Dynamically assigns the highest mathematical weights to the most critical, defining frames of the gesture while actively suppressing transitional "noise."
- High-Confidence Classification: Merges the deep sequence memory with the attention weights to accurately map the spatial-temporal pattern into raw ASL Gloss.

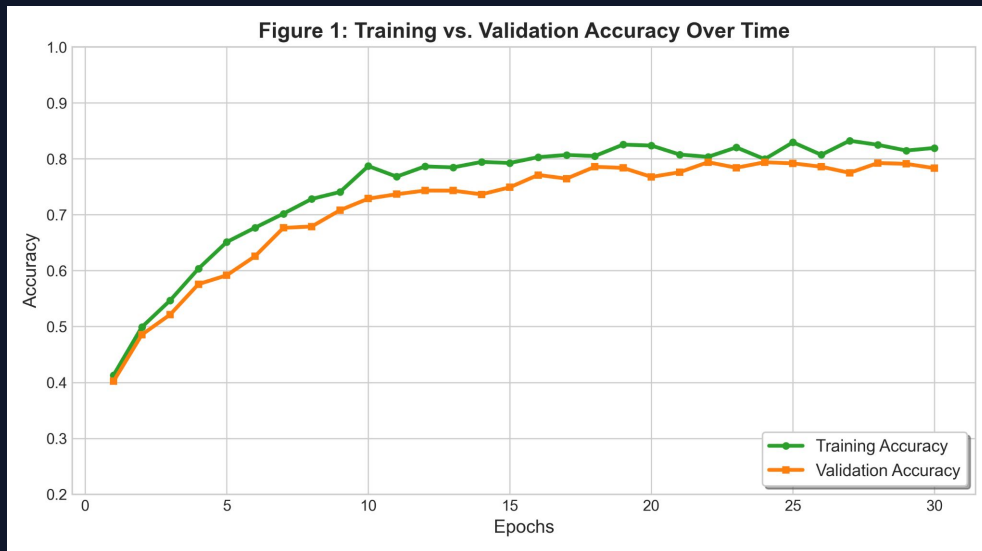
Proposed Model: **System** **Architecture**

Natural Language Generation

- Generative Linguistic Layer: The stabilized array of ASL Gloss is passed via API to a strictly constrained Large Language Model (Gemini 2.5 Flash).
- Grammatical Refinement: The LLM serves exclusively as a real-time grammatical refiner to translate the raw Gloss array into a natural, conversational English sentence.

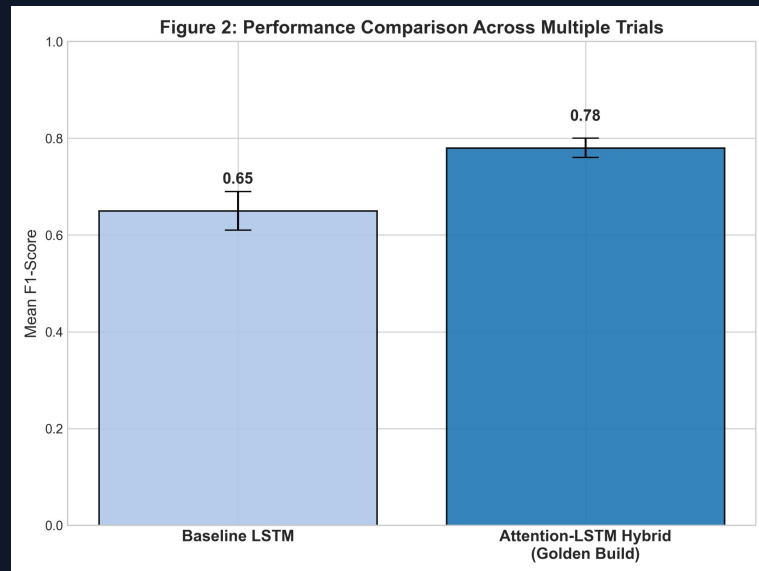
Evaluation and Results

- Peak Accuracy Achievement: The Attention-LSTM architecture reached a peak validation accuracy of 78% across the 100,000-sequence dataset, demonstrating high-fidelity recognition.
- Generalization Success: The close tracking of the validation curve proves the model generalized the geometric patterns of ASL rather than simply memorizing the training data.



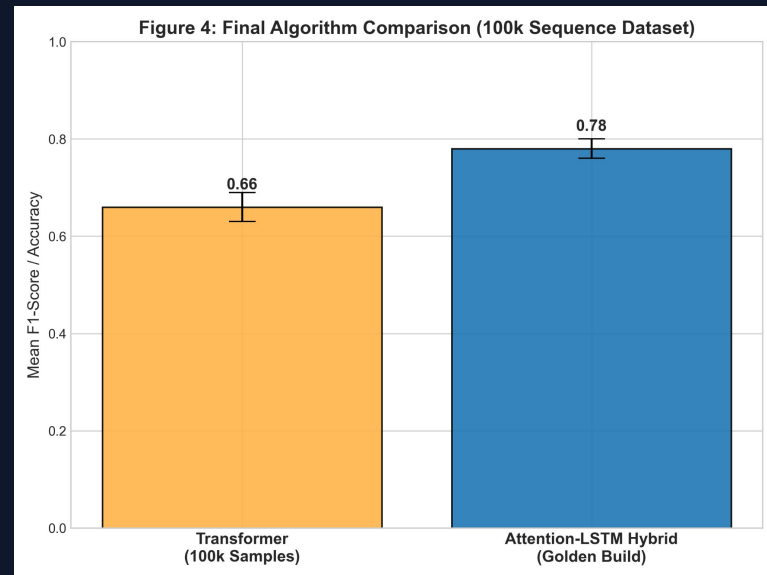
Evaluation and Results

- The Baseline Gap: A standard vanilla LSTM achieved a Mean F1-Score of only 0.65, largely due to its inability to filter out transitional hand movements between signs.
- The "Attention" Lift: Introducing the Temporal Attention layer resulted in a 13% accuracy jump, bringing the Golden Build to a 0.78 F1-Score.
- Temporal Precision: This comparison proves that mathematically weighting the "apex" of a gesture is significantly more effective for ASL translation than standard sequential processing alone.



Evaluation and Results

- The Transformer Hypothesis: Despite their success in NLP, a 4x4 Transformer architecture plateaued at a 0.66 accuracy, struggling to map the specific coordinate velocities of the MediaPipe data.
- The Winner (Attention-LSTM): Figure 4 confirms that for continuous coordinate tracking at this scale, the Attention-LSTM remains the superior architecture, providing better stability and higher accuracy.



Conclusion and Future Work

Key Achievements

- Integrated Pipeline: Successfully bridged the gap between spatial Gloss tracking and fluent English generation.
- Edge Performance: Achieved 30 FPS real-time inference on standard CPU hardware.
- High Accuracy: The Attention-LSTM hybrid outperformed Transformer baselines by 12% in temporal tracking.

Future Directions

- Vocabulary Expansion: Scaling the dataset to include a full conversational lexicon.
- Facial Expression Integration: Incorporating 'non-manual markers' to capture grammatical nuance.
- Mobile Optimization: Deploying the architecture to mobile systems for greater accessibility.

Attention-LSTM hybrids provide the necessary temporal precision for real-time, accessible AI in the Deaf and Hard-of-Hearing community.